

BIF401-Bioinformatics-I

MCQs past

MID TERM-QUIZ NO 1

- 1. FASTA achieves alignment by using **short lengths of exact matches**.
- 2. Dot plots cannot capture **insertions, deletions and gap indications**
- 3. Dot plots employ dot matrix with two sequences plotted on **top and left of the matrix**
- 4. In multiple sequencing, we have **both**
- 5. CLUSTALW can run in two modes **slow/accurate, fast/appropriate**.
- 6. The correct order of progressive alignment for multiple sequencing alignment is **Pair wise alignment, Guide tree, branching order in tree**.
- 7. Blast can find sequencing from protein or nucleotide database **it can find known and unknown sequencing**.
- 8. **: Next generation sequencing** is used to known about unknown sequences of nucleotide.
- 9. In multiple sequence algorithm **doing so many progressive alignment can be slow**.
- 10. : Structural information of a molecule obtained by **X ray crystallography & Nuclear magnetic resonance**.
- 11. Before starting to mach nucleotide or amino acid are need to find the **Location of conserved residues**.
- 12. Bioinformatics is an interdisciplinary relative new development of **biology, mathematics, computer science, chemistry and physics**.
- 13. Fasta online software Can also perform genome and proteome similarity search .**true**
- 14. From Fasta we can obtain the information **on the parent organism, function and evolution history**.
- 15. name any two frequently used scoring matrix **Blosum and pam**.
- 16. Amino acids are chained together by **peptide bonds**
- 17. To find out the unknown sequence of nucleotide and protein we use **blast**.
- 18. GenBank and Uniprot, have **A large number of sequences are available in publically** .
- 19. Protein databank is a **data base**
- 20. In dynamic programming we can do multiple sequence alignment upto **21 sequence** .
- 21. How many types of secondary structure can be formed possible of RNA? **5**.
- 22. RNA structures have the **lowest (or close)** quantity of free energy.
- 23. RNA can play role due to its structure . **all of above**.
- 24. Rna contains **ribose** sugar?
- 25. Thymine is replaced by **uracil** in rna.
- 26. RNA molecules readily degrade due to their **short** shelf life .
- 27. Each Nucleotide in rna has **covalently bonded** to the backbone .
- 28. **Gibbs Free Energy”** is the free energy of an RNA molecule available for reaction .
- 29. The structure of rna sampled can be measured by.**Atomic Force Microscopy**
- 30. The 3’-end of **mRNA** has a poly A tail
- 31. The role of **mRNA** Carry genetic information from DNA to Ribosomes where proteins are being assembled .
- 32. Messenger rna is only **5-10%** of rna content in the cell.

FOR MID TERM PREPARATION

VU APEX CAMPUS	vuapex.com.pk	vuapex.pk
Contact Us:	0322-8877744	

- 1: Gaps are produced by **insertion or deletion**.
- 2: Example of energy base **Zuker algorithm**.
- 3: **Two** forms of DNA rending sequence.
- 4: Translation involves coding of protein by **RNA** at ribosome.
- 5: some medicine have side effect on certain patient **Frontiers in personalized medicine**.
- 6: By comparing sequence we can get **similarity, specific difference, relationship, evolutionary insight**.
- 7: Non clustering method include **Maximum parsimony**.
- 8: Destabilizing energy of a nucleotide has **(+) value**.
- 9: Comparison of protein sequence can be done with...**Uniprot**
- 10: Function is a property of the RNA structure, which is formed from the **RNA sequences**.
- 11: Structural information of a molecule obtained by **resonance**. **X ray crystallography & Nuclear magnetic**
- 12: Protein have **20** amino acid.
- 13: If sequence are 9 then according formula it will be **9²**.
- 14: Fast is **available online**.
- 15 Fasta was developed in **1988**.
- 16: Bioinformatics is the inter disciplinary relative new **developing field**.
- 17: Genbank can be reached by...online website or www.ncbi.nlm.nih.gov/genbank.
- 18: Before starting to mach nucleotide or amino acid are need to find the **Location of conserved residues**.
- 19: some medicine have side effect on certain patient **Frontiers in personalized medicine**.
- 20: Identity is the number of **nucleotide or amino acid** which match exactly between two biological sequence.
- 21: **Alignment** are represented by diagonals in dot matrix.
- 22: Needleman Wunsch algorithm we use for..... **Global alignment**.
- 23: **Exons** in protein tend to be more conserved then the **intron**.
- 24: In MSA the distance between 2 sequences in equal to **NO of identical residue/number of aligned residue**.
- 25: All the changes which accrued in our cell **metabolism**
- 26: The loop of the hairpin must be least **4 bases** lo
- 28: NJ algorithm is used to produc **Optimal RNA 2' sturcture**.
- 29: **energy base** algorithm type is Zuker algorithm.

VU APEX CAMPUS	vuapex.com.pk	vuapex.pk
Contact Us:	0322-8877744	

- 30: Types of RNA...present in organism. (5)
- 31: The online search is for DNA and RNA is..... genebank 32:
- Rooted tree is computational expensive.
- 33: In multiple sequence algorithm doing so many progressive alignment can be slow.
- 34: DNA dictates the production of cell protein and carbohydrate
- 35: protein have long sequences ...in no?
- 36: In Exact partial matching of residue and order with room for variation in both.
- 37: DNA/RNA sequence is available on.... GENE BANK 38:
- Martine algorithm predict for structure 2 RNA.
- 39: In extraction of ORF, 1st step is obtained the 3 codons.
- 40: Example of energy based algorithm Zukar algorithm.
- 41: Dynamic programming help us to..... reduce computational cost.
- 43: When unpaired bases in 2 region binds is result form Buldges.
- 44: which one of following is not objective based method of phylogenic ... UPGMA 45:
- protein sequencing can be done by Swiss prot & UniProt.
- 46: Drugs can be better developed after understanding molecular basis of Disease
- 47: Online tools are available for homology modeling.
- 48: Insertion and deletion in alignment cause..... gaps.
- 49: Alligment of insertion and deletion can be performed by dynamic programming
- 50: Exons in proteins tend to be more conserved than intron.
- 51: Blast confined sequences from—— ID,name,spcies
- 52: ribosmes read RNA and collect amino acid from..... cell cytosol.
- 53: Simulation can predict disease progression & outcomes.
- 54: For exact machingrequired. Traceback 55: traceback
- is start from high score on couple nucleotide.
- 56: Motif protein chain fold to 3D structure take .
- 57: Next generation sequencing is used to known about unknown sequences of nucleotide.
- 58: RNA fold to form 2 structure.
- 59: m-odd implement are ungraded version of Zuker.
- 60: Which has less computational cost Dynamic programming
- 61: In scoring matrices, we with maximum 4 score matrix.

VU APEX CAMPUS	vuapex.com.pk	vuapex.pk
Contact Us:	0322-8877744	

- 62: We compare all length of nucleotides in **Global alignment.**
- 63: we compare smaller fragment **Local alignment.**
- 64: Destabilized RNA molecule have **(+)value-**
- 65: In Zukar algorithm, we need construct **all possible combination.**
- 66: We use **Zukar algorithm** to sum up energy to find lowest matrix.
- 67: **lowest energy** is selected in Zukar algorithm.
- 68: In comparing RNA structure, we used to select which having **lowest energy.**
- 69: Addition of nucleotide is done by **hydrogen bond**
- 71: Adman degradation is used for **protein sequencing.**
- 72: Why do we create all possible combination of diagonals in zukar algorithm with overall energy is selected.
- 73: We can do repeated sequences alignment by slight modification in **waterman model.**
- 74: Length of hairpin loop is **4 bases.**
- 75: "T" Threonine is an basic element....**true**
- 76: To align sequence what will be necessary **to find conserved region** firstly.....
- 77: Sequence of biomolecules can be known by **uniprot or gene bank** using
- 78: **ExPASy** provides access to variety of online databases and tools.
- 79: Structural information of a molecule can be obtained by **X-ray crystallography , nuclear magnetic resonance.**
- 80: Compare a protein query sequence against a nucleotide sequence database.....**BLAST**
- 81: Protein formation of abnormal folding called **Pseudoknots.**
- 82: By using uniprot we can find sequence of.....(**protein**)
- 83: The 5' end of molecule capped with **7 Methyl Guanosine Triphosphate.**
- 84: **DP** provides base pairing for binding.
- 85: DNA get modified by **mutation, substitution, insertion, deletion.**
- 86: UPGMA is used to calculate the **distance of each cluster.**
- 87: Global alignment used to determine overall conversation.
- 88: if the sequence is multiple there can be several possible alignment.
- 89: can we identify the post translational **ExPASy** modifications by .
- 90: scoring scheme has an extremely **crucial** role in producing **optical** alignment.
- 91: why is there need to predict RNA structures **to know about its function**
- 92: DP recombines the nucleotide recombination through a process of **Traceback.**
- 93: .how many open reading frame exist for each DNA sequence.**6ORFs**

- 94: tblastn search translated nucleotide using... **Protein query**
- 95: Blast can find sequencing from protein or nucleotide database **it can find known and unknown sequencing.**
- 96: name any two frequently used scoring matrix **Blosum and pam.**
- 97: From Fasta we can obtain the information on **the parent organism, function and evolution history.**
- 98: The correct order of progressive alignment for multiple sequencing alignment is **Pair wise alignment, Guide tree, branching order in tree.**
- 99: CLUSTALW can run in two modes **slow/accurate, fast/appropriate.**
- 100: Gene bank and UniProt have **Large number of sequencing publically.**
- 101: In multiple sequencing, we have **Global alignment.**
- 102: Ubiquitin play important role recycling **protein.**
- 103: **6** form of DNA ending sequencing.
- 104: Non-clustering method such as **maximum parsimony** may be used for making trees.
- 105: If in needleman and Smith watermark algorithm along '0' place is in.....relationship. **1st row and 1st column.**
- 107: Faster active alignment by using local alignment and protein nucleotide. **pairwise alignment**
- 109: Expasy was develop by **Swiss Bioinformatics Institute (SIB)**
- 110: .In phylogenetics tree , ancestor are placed at root bottom and sister node trees.
- 111: .In UPQMA two sequence **N_x & N_y** are .
- 112: .When unpaired bases in **2** region binds itself form **Buldges.**
- 113: **Zukers algorithm** is use to predict computational and stabilizing energy?
- 114: In Nussinov Jacob algorithm which turn is added to compare to score for diagonal position set **diagonal and lower tri diagonal to zero.**
- 115: Trackback begins from the position in Nussinov Jacobson score matrix. **4rth possition**

Genomics, evolutionary studies and system biology are application of **bioinformatics.**

Exact matching require which type of nucleotide-**same number of nucleotide.**

Accurate solution of protein structure is one of **the toughest problems** **Tract**

back is overlap matches can start from any position in scoring matrix.

The matrix hasscores. **Positive and negative.**

VU APEX CAMPUS	vuapex.com.pk	vuapex.pk
Contact Us:	0322-8877744	

MASCOT....._can search people mass finger printing and shotgun proteomics data set

7) Spectrometer measures the protein by_____ **Mass/charge**

Simulation of the folding process depends_____ **Energy function**

Information about **DNA/RNA** is available on gene bank

VU APEX CAMPUS	vuapex.com.pk	vuapex.pk
Contact Us:	0322-8877744	